MIT SCHOOL OF COMPUTING
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MIT-ADT UNIVERSITY
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A Leap Towards World Class Education

Name: Viraj M. Bhosale Class: SOC-9 Batch: C
Subject: PL-I (Python) Roll No.: 61. Exp. No.: 1
Name of the Experiment: Introduction to python programming.
Performed on:
Submitted on:

Marks	Teacher's Signature with date
10	

1) List the key features & advantages of python. Also mention any two python IDE's or editors used for program developement.

Key features →

1) Simple and readable syntax.
2) High level and a interpreted langage.
3) platform independent
4) Object oriented.
5) Large std library.

* Advantages

1) Easy to learn.
2) Faster developement.
3) less code.
4) Widely used in Industry.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

1) Explain the Industry history and evolution of python describe why phython has become popular for beginners and Industry applications

- python was developed by Guide Van Rossum. in 1989 and realesed in 1991
- Evolution of python.

python 1.0 → Basic features and simple syntax.
python 2.0 → Added list comprehesions and garbage collection.
python → 3.0 → Improved unicode support and better syntax also it is the current version.
= Reason for python's popularity

i) Easy to learn and understood
2) Used in Industry and academics.
3) Strong support for AI, ML & DS.
4) Huge history library ecosystem.

3) 2) Write a programm to accept two numerical values and perform arithmetic operations.

→ # Accept two numbers
a= int (input ("Enter first number :"))
b= int (input ("Enter second number:"))
print ("Addition :")) (a+b).
print ("Subtraction :" a-b).
print ("Multiplication :" a×b))
print ("Division :," a/b).

Q.3) Write a python program to accept two numeric values from the user and the perform the following questions.

→ Addition, Subtraction, multiplication, Division, Display results, using proper input and output prompting.

→

• Addition

```
num 1=2
num 2=4
Sum num 1+ num2
print (Sum)
```

output :6

• Subtraction.

```
num1=5.
num 2=3
Sub =num1 -num2.
print (Sub).
```

output : 2.

4) Analyze the following python code & identify the data type of each variable. Also explain the output.

→

```
a=10
b=5.5
c= 3+4j
d= True.
print (a +b, type (c), d).
```

output -

Q.5) Evaluate the use of relational logical (and) bitwise operators in python. Justify with suitable examples where each types of operators is most appropriate.

① Relational operators >, <, >=, <=, ==, !=
use to compare 2 values.

Eg age = 18
if age >= 18:
print (" Eligible to vote").
else :
print (" Not eligible") # True.

② logical operator - and, or, Not.
- more than 1 condition to check.

Eg. username = " admin".
password = "1234".
If username = " admin".
print (" logic successfull") # True.
else:
print (" Acess denied").

③ Bitwise operators = & | ^ ~, <<, >>
- Performance optimization.

Eg. a = 5
b = 3.
result = a & b.
print (result).
output : 1.

Data type of variables.

a = 10 - Integer
b = 5.5 - Float.
c = 3+4j - Complex.
d = True - Boolean.

output

1) 15.5 < class 'complex' > True.
2) a + b = 10 + 5.5 = 15.5.
   Integer + Float = Float.

3) Type (c).
   'c' is a complex number.

4) 'd'.
   Boolean value is True is printed.

Print (a+b, type (c), d).

15.5 < class 'complex' > True.

Q.6) Design and Implement a menu-driven python program that →.

→ Accept user input.

→ performs arithmetic, relational, logical and assignment operations.

→ Display results clearly.

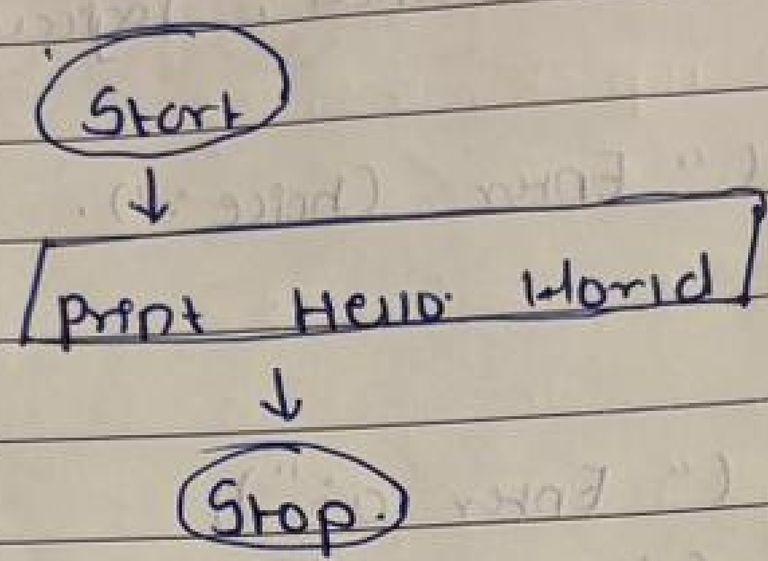
→ True.

```
print (Arithmetic, Relational, logical, Assignment
, Exit\n").
Ch = int (input (" Enter Choice :").
if
 Ch = 1;
 (a = float (input (" Enter a:"))
 (b = float (input (" Enter b:")).
 (print (" add :", (a+b).

else if Ch = 2.
 (a = int (input (" Enter a :")).
 (b = int (input (" Enter b:"))
 print (" a>b :" a>b)

elif Ch = 3.
 a = int (input (" Enter a :").
 b = int ( input (" Enter b:").
 print (" a>b:", a>b).

elif ch = 4.
 Break
else'
 print (Invalid choices").
```

Q.7) Write a program in python to display "Hello World".

→

Step 1: Start.
Step 2: Display the message "Hello world".
Step 3: Stop.



Code- print ("Hello World").

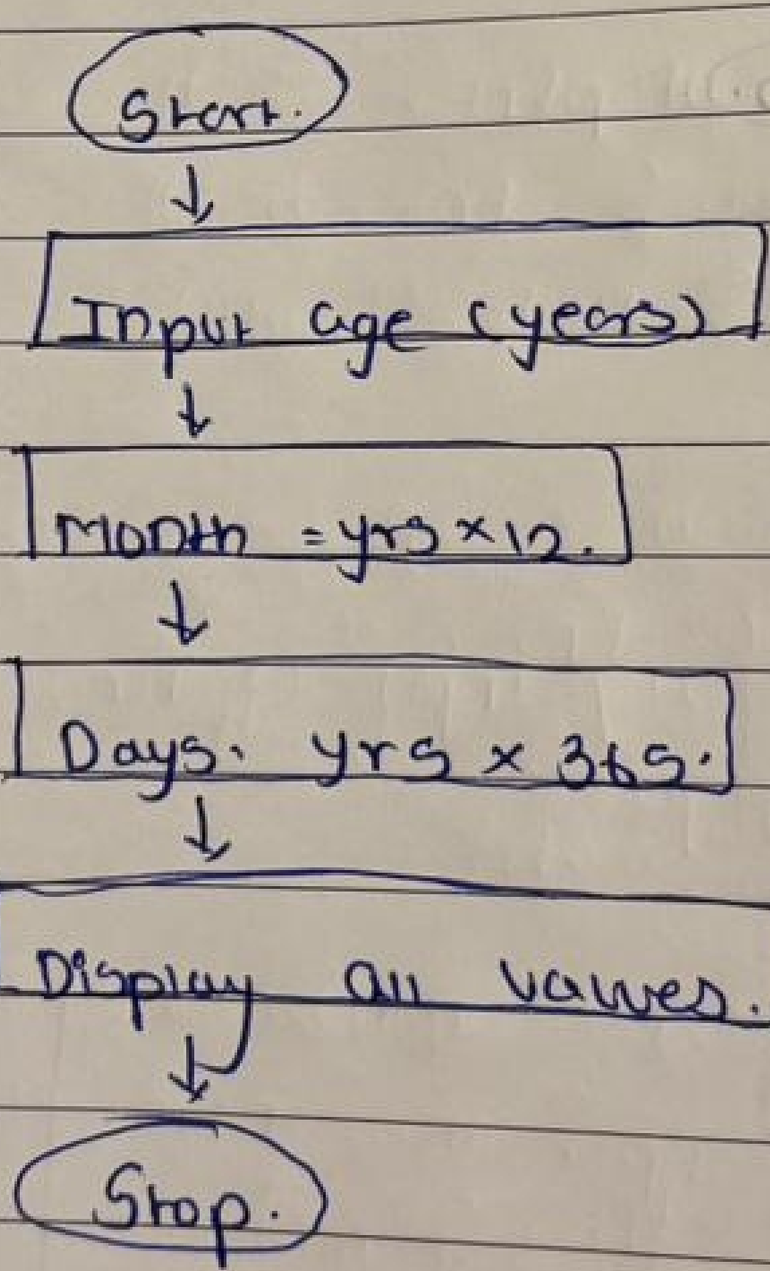
Q.8) Write of program to perform average of five number using input () function and type conversion concept.

→ Step 1 = Start.
Step 2. Read five numbers using input().
Step 3. Convert input into numeric value.
Step 4: Add all five number.
Step 5: Divide the sum by 5 to get avg.
Step 6 = Display the avg.
Step 7: Stop.

Q.9) Write a program that prompt the user to enter thier age and print out thier age in years, month and days.

→

Step 1: Start.
Step 2: Read age in years.
Step 3: Calculate month = yrs × 12.
Step 4: Calculate days: yrs × 365.
Step 5: Display age in yrs, month and days
Step 6: Stop.



```
a = int (input (" Enter an integer value :")).
b = float (input (" Enter a float valued)).
c = input (" Enter a number as string":) :
print (" Before Conversion).
print (" a = " a, 'Type :" type (a)).
print ( b = " b, 'Type :" type (b)).
c-int = int (c).
sum - value = at. (c).
print (" In After Conversion").
print(" c - int = " c-int, 'Type, "Type( (c-int)).
print (" sum of 'a' & c-int = ", sum -value).
```

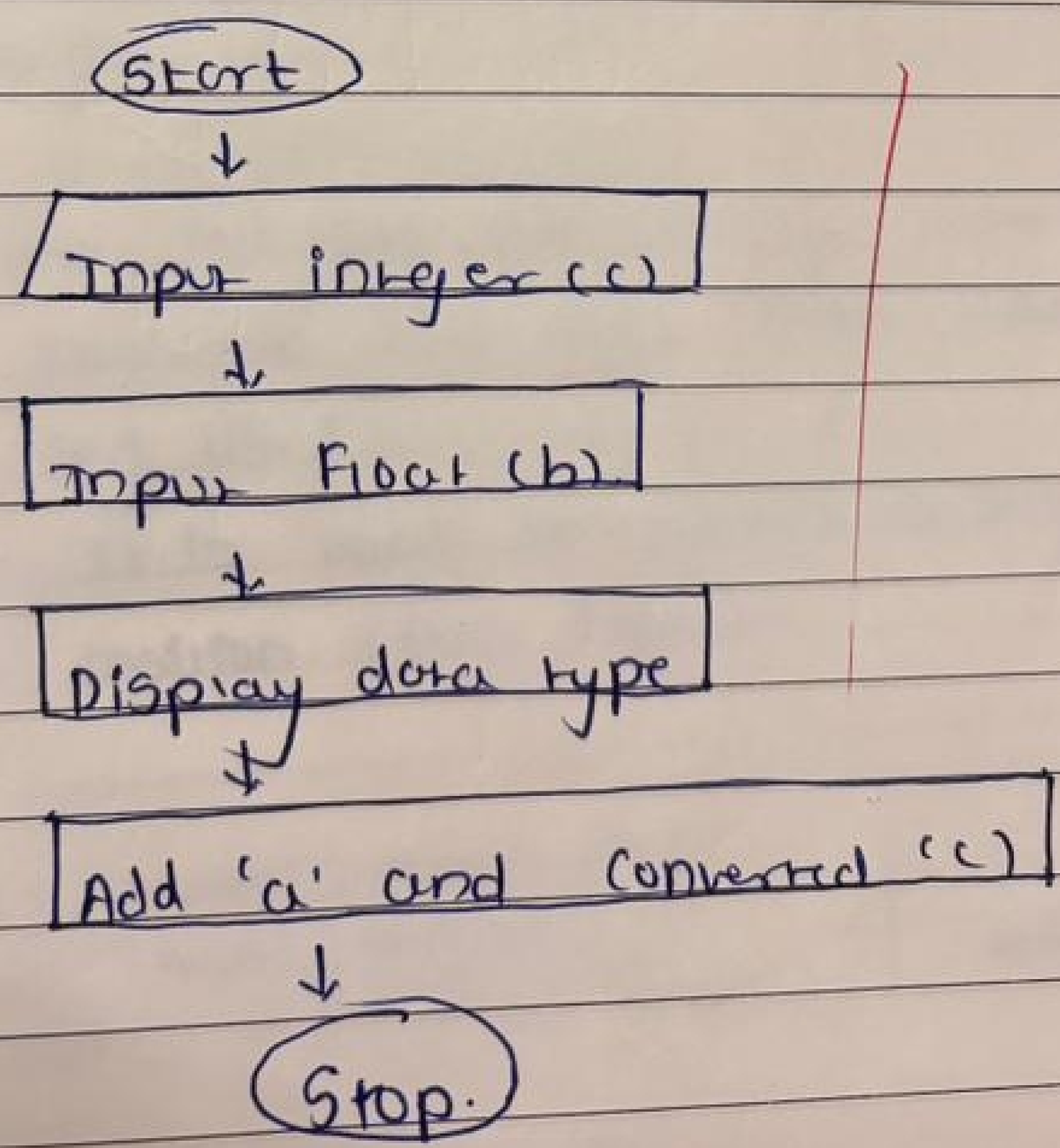
Q.9) → age - years = int (input (" Enter your age in years:")
age - months = age - years * 12.
age - days = age - years ", age - years).
print ("age in years", agen - years).
print (" age in months", age - months).
print (" age in Days :", age - days).

Q10) Write a program to demonstrate the concept of data type and conversion.

1) Start.
2) Input an Int value.
3) Input a Float value.
4) Input a No number as String.
5) Display the data type of each.
6) Convert String value into Integer.
7) Stop.

~~Algo~~:-

* Flow Chart :-



Start

↓

[Input 5 Number using input]

↓

[Convert input to Float values.]

↓

[Sum = n1 + n2 + n3 + n4 + n5.]

↓

[Avg = Sum / 5]

↓

[Display Avg]

↓

Stop.